

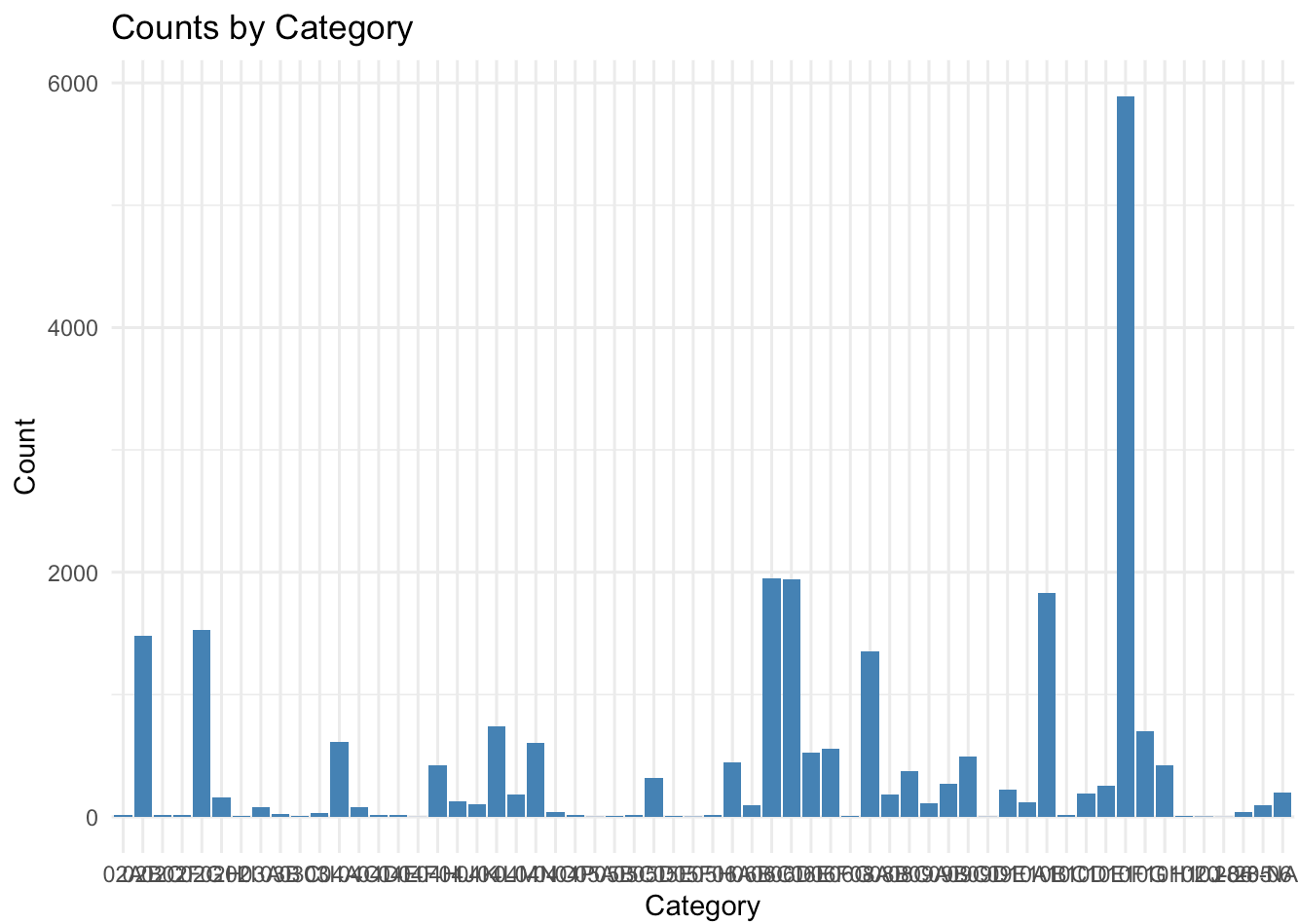
HDL Visualizations

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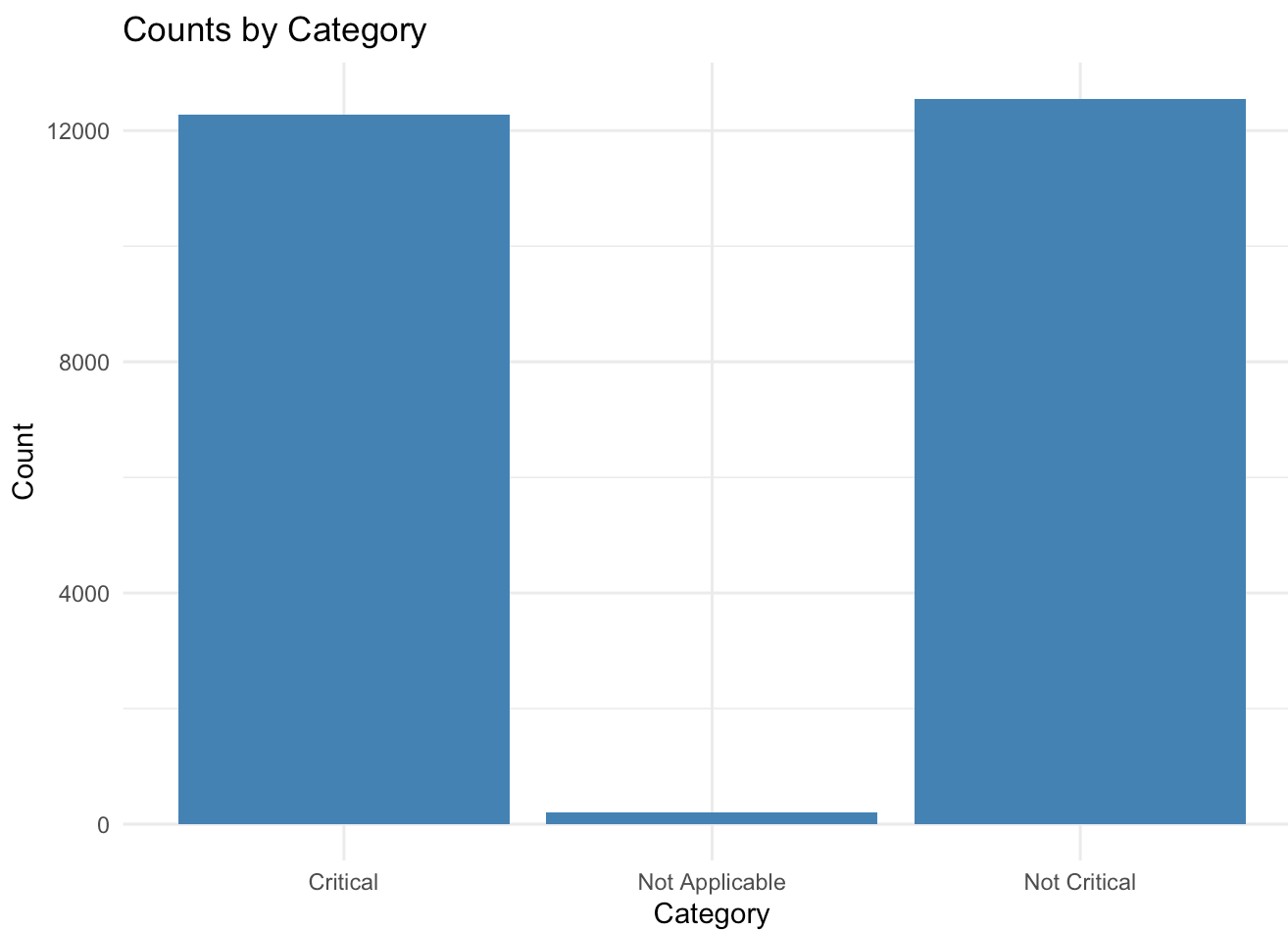
2026-07-11

Plots

```
#violation code plot
ggplot(restaurant_data, aes(x = violation_code)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Counts by Category",
    x = "Category",
    y = "Count"
  ) +
  theme_minimal()
```

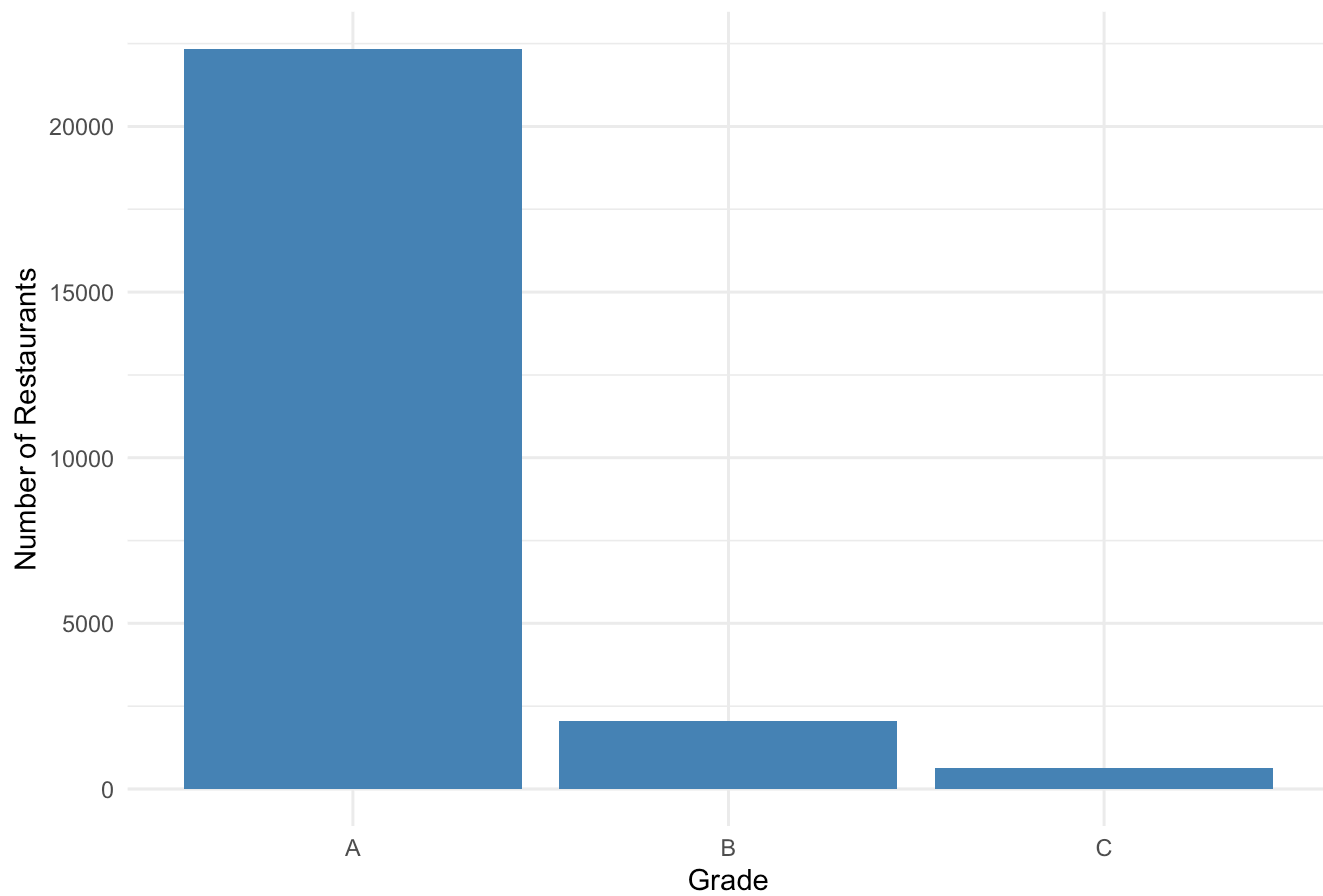


```
#critical/not critical plot
ggplot(restaurant_data, aes(x = critical_flag)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Counts by Category",
    x = "Category",
    y = "Count"
  ) +
  theme_minimal()
```



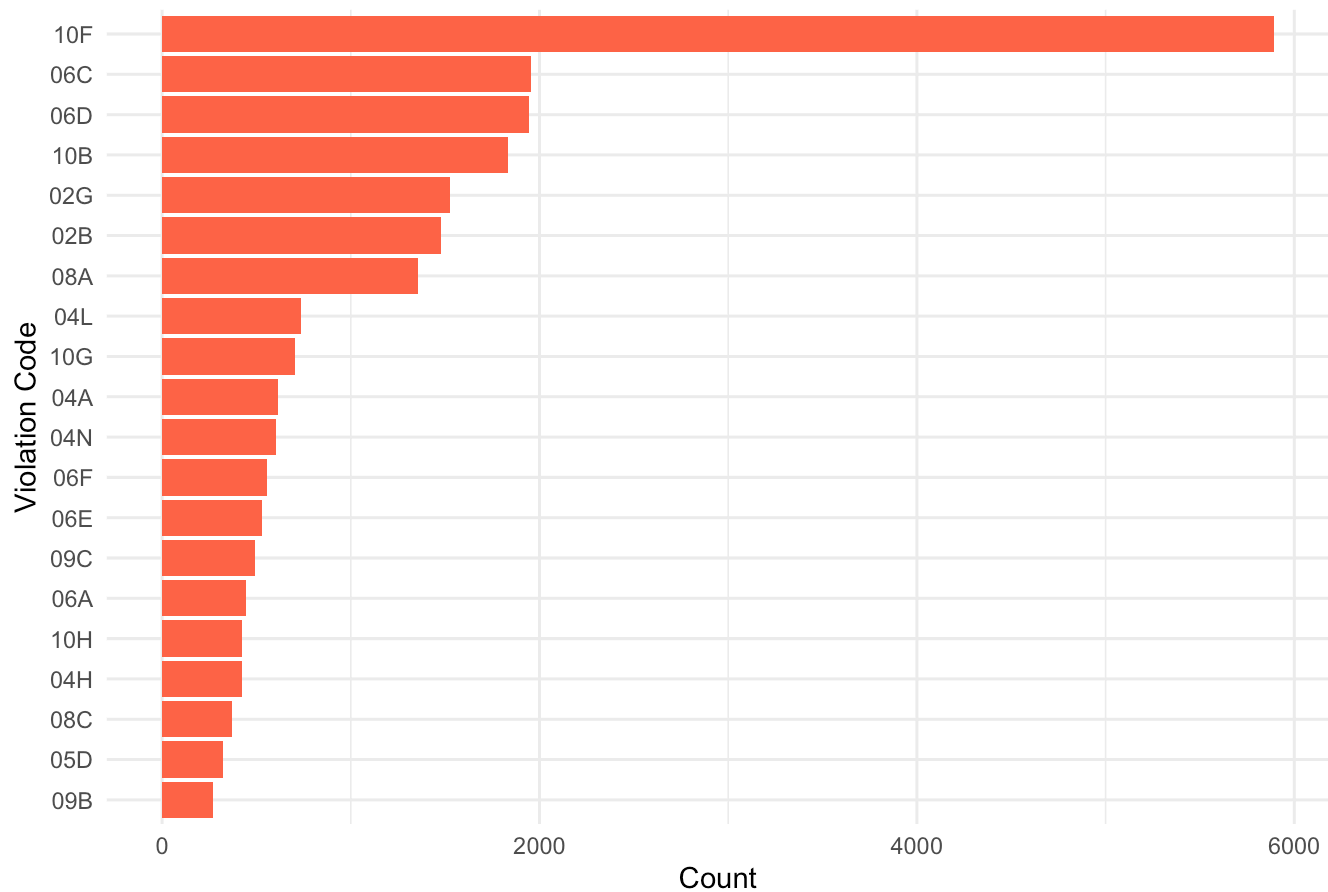
```
#restaurant grade plot
ggplot(restaurant_data, aes(x = grade)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Restaurant Inspection Grades",
    x = "Grade",
    y = "Number of Restaurants"
  ) +
  theme_minimal()
```

Restaurant Inspection Grades



```
#most common violation codes
restaurant_data %>%
  count(violation_code, sort = TRUE) %>%
  slice_head(n = 20) %>%
  ggplot(aes(x = reorder(violation_code, n), y = n)) +
  geom_col(fill = "tomato") +
  coord_flip() +
  labs(
    title = "Top 20 Most Common Violation Codes",
    x = "Violation Code",
    y = "Count"
  ) +
  theme_minimal()
```

Top 20 Most Common Violation Codes



#most common violation descriptions

```
restaurant_data %>%
  count(violation_description, sort = TRUE) %>%
  slice_head(n = 15) %>%
  ggplot(aes(x = reorder(violation_description, n), y = n)) +
  geom_col(fill = "darkorange") +
  coord_flip() +
  labs(
    title = "Top 15 Violation Descriptions",
    x = "Violation Description",
    y = "Count"
  ) +
  theme_minimal()
```

intact surface or equipment made of unacceptable material, not kept clean, or not properly sealed, raised, spaced or movable to allow for proper cleaning; Food contact surface not properly washed, rinsed and sanitized after each use; contamination during storage, preparation, transportation, display, service or from customer's refillable, reusable container. Condiment container not properly sealed; hand sanitizer not provided where required; equipment or floor not properly drained; sewage disposal system in disrepair or not functioning; food held at or above 41 °F; hot food held above 135 °F; cold food held below 41 °F; intact raw eggs held above 45 °F; or reduced oxygen packaged (ROP) TCS foods held at or above 38 °F; intact raw fish held above 40 °F; or intact raw poultry held above 40 °F.

Establishment is not free of pests.

Dishwashing and ware washing: Cleaning and sanitizing procedures not followed.

Food Protection

Food stored in establishment's food and/or non-food areas. FRSA flies include house flies, fruit flies, and other nuisance pests associated with (FRSA) flies or other nuisance pests in establishment's food and/or non-food areas. FRSA flies include house flies, fruit flies, and other nuisance pests.

Wiping cloths not properly sanitized.

Sanitized equipment or utensils not properly stored.

Design, construction, materials used or maintenance of food contact surfaces not adequate. Outer garment soiled with possible contaminant. Effective hair restraint not worn where required. Jewelry worn on hands or wrists.

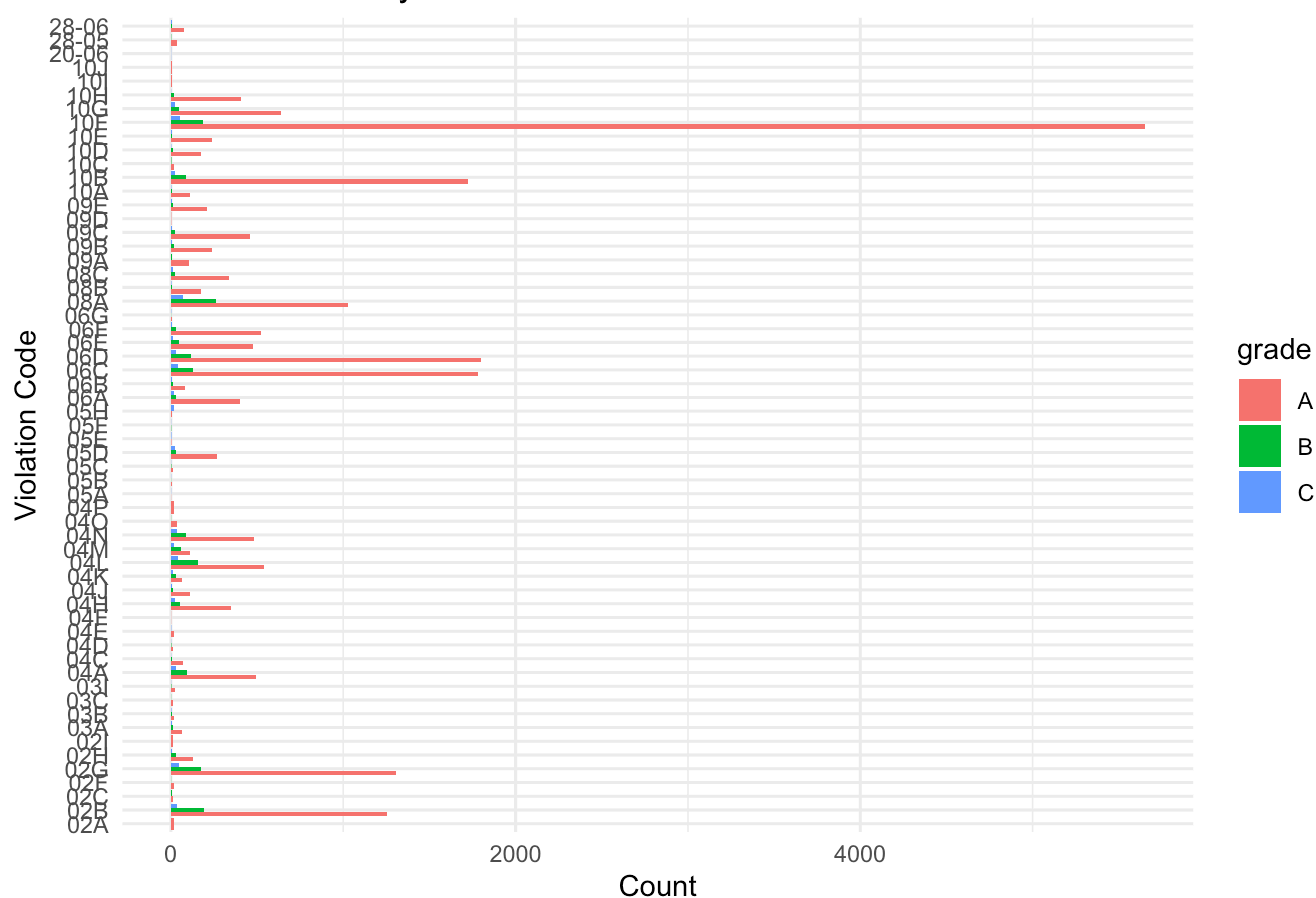
#violation codes & descriptions

```
View(
  restaurant_data %>%
    distinct(violation_code, violation_description) %>%
    arrange(violation_code)
)
```

#code and grade

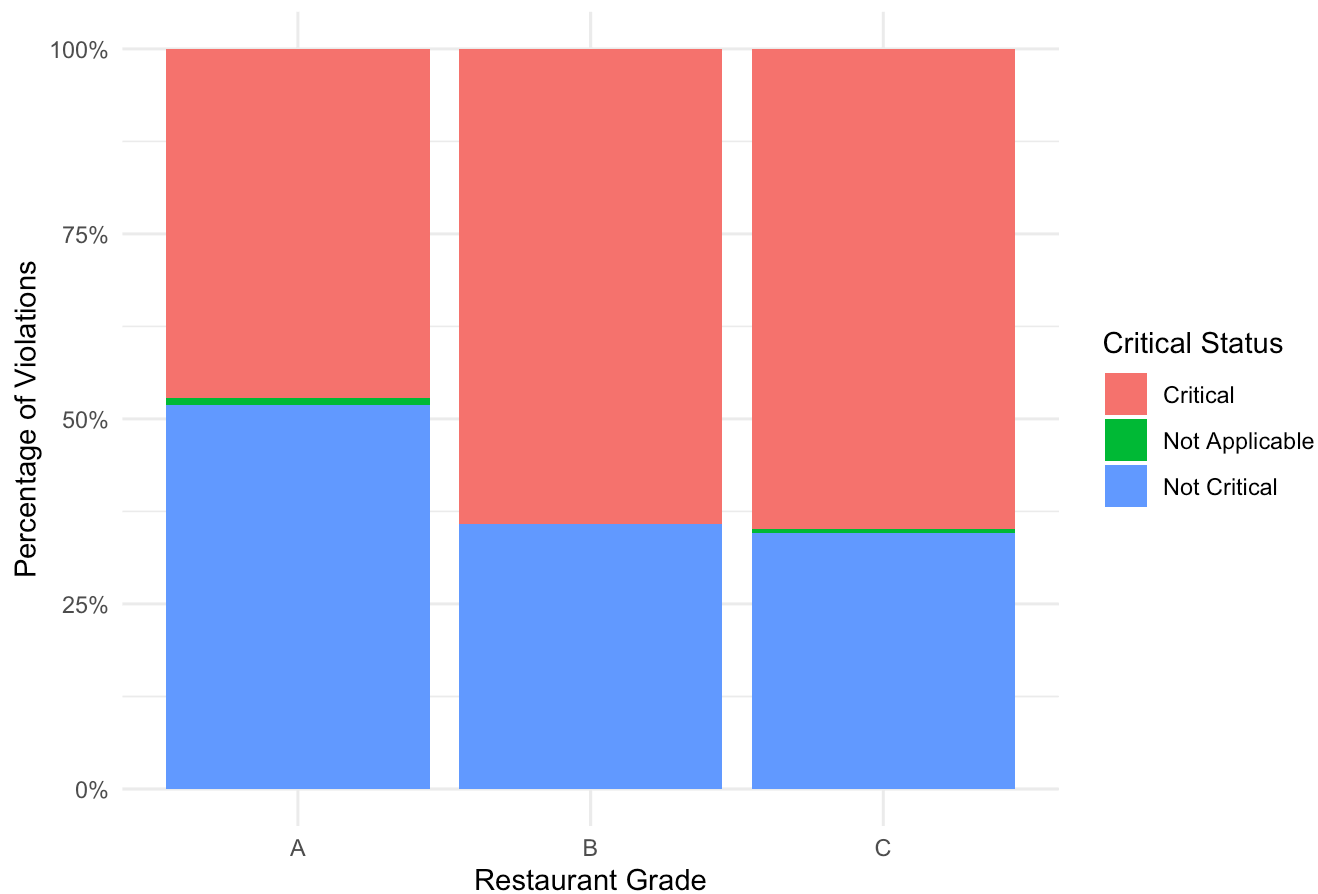
```
restaurant_data %>%
  filter(!is.na(grade), !is.na(violation_code)) %>%
  count(grade, violation_code) %>%
  ggplot(aes(x = violation_code, y = n, fill = grade)) +
  geom_col(position = "dodge") +
  coord_flip() +
  labs(
    title = "Violation Codes by Restaurant Grade",
    x = "Violation Code",
    y = "Count"
  ) +
  theme_minimal()
```

Violation Codes by Restaurant Grade



```
#grade vs critical
restaurant_data %>%
  filter(
    !is.na(grade),
    !is.na(critical_flag),
    grade %in% c("A", "B", "C")
  ) %>%
  ggplot(aes(
    x = grade,
    fill = critical_flag
  )) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent) +
  labs(
    title = "Critical Violations by Restaurant Grade",
    x = "Restaurant Grade",
    y = "Percentage of Violations",
    fill = "Critical Status"
  ) +
  theme_minimal()
```

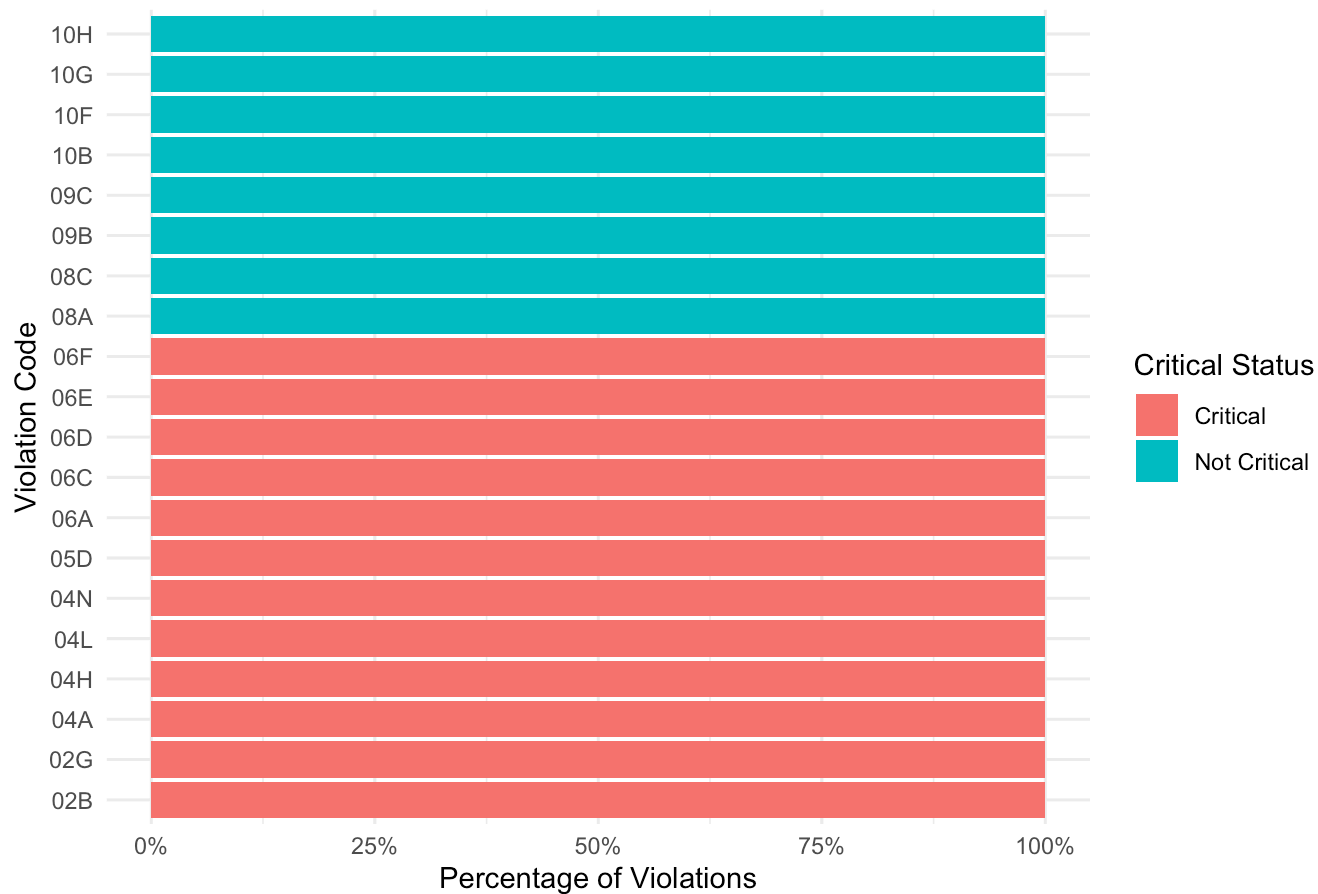
Critical Violations by Restaurant Grade



```
# Find the 20 most common violation codes
top_codes <- restaurant_data %>%
  filter(!is.na(violation_code)) %>%
  count(violation_code, sort = TRUE) %>%
  slice_head(n = 20) %>%
  pull(violation_code)

# Graph critical status by violation code
restaurant_data %>%
  filter(
    violation_code %in% top_codes,
    !is.na(critical_flag)
  ) %>%
  ggplot(aes(
    x = violation_code,
    fill = critical_flag
  )) +
  geom_bar(position = "fill") +
  scale_y_continuous(
    labels = function(x) paste0(round(x * 100), "%")
  ) +
  coord_flip() +
  labs(
    title = "Critical Status of the 20 Most Common Violation Codes",
    x = "Violation Code",
    y = "Percentage of Violations",
    fill = "Critical Status"
  ) +
  theme_minimal()
```

Critical Status of the 20 Most Common Violation Codes



Summaries

```
#grade count & percentages
```

```
restaurant_data %>%  
  count(grade) %>%  
  mutate(percent = round(n / sum(n) * 100, 2))
```

```
## # A tibble: 3 × 3  
##   grade      n percent  
##   <chr> <int>   <dbl>  
## 1 A      22343   89.3  
## 2 B       2047    8.18  
## 3 C        636    2.54
```

```
#violation code count
```

```
restaurant_data %>%  
  count(violation_code, sort = TRUE)
```



```
## # A tibble: 60 × 2
##   violation_code      n
##   <chr>          <int>
## 1 10F             5891
## 2 06C             1952
## 3 06D             1944
## 4 10B             1834
## 5 02G             1527
## 6 02B             1478
## 7 08A             1356
## 8 04L              738
## 9 10G              702
## 10 04A             614
## # i 50 more rows
```

```
#10 most common violations
restaurant_data %>%
  count(violation_code, violation_description, sort = TRUE) %>%
  slice_head(n = 10)
```

```
## # A tibble: 10 × 3
##   violation_code violation_description      n
##   <chr>          <chr>                <int>
## 1 10F            Non-food contact surface or equipment made of unaccepta... 5819
## 2 06D            Food contact surface not properly washed, rinsed and sa... 1944
## 3 06C            Food, supplies, or equipment not protected from potenti... 1918
## 4 10B            Anti-siphonage or back-flow prevention device not provi... 1822
## 5 02G            Cold TCS food item held above 41 °F; smoked or processe... 1519
## 6 02B            Hot TCS food item not held at or above 140 °F.             1468
## 7 08A            Establishment is not free of harborage or conditions co... 1323
## 8 04L            Evidence of mice or live mice in establishment's food o...  727
## 9 10G            Dishwashing and ware washing: Cleaning and sanitizing o...  685
## 10 04A           Food Protection Certificate (FPC) not held by manager o...  597
```

```
#num of unique violations
restaurant_data %>%
  summarise(
    unique_violation_codes = n_distinct(violation_code),
    unique_descriptions = n_distinct(violation_description)
  )
```

```
## # A tibble: 1 × 2
##   unique_violation_codes unique_descriptions
##   <int>                <int>
## 1           60          101
```

```
#grade by violation
restaurant_data %>%
  group_by(grade) %>%
  summarise(
    inspections = n(),
    unique_violations = n_distinct(violation_code)
  )
```

```
## # A tibble: 3 × 3
##   grade inspections unique_violations
##   <chr>      <int>          <int>
## 1 A         22343             58
## 2 B         2047             47
## 3 C          636             46
```

```
#most common violation by grade
restaurant_data %>%
  group_by(grade, violation_code, violation_description) %>%
  summarise(count = n(), .groups = "drop") %>%
  arrange(grade, desc(count))
```

```
## # A tibble: 219 × 4
##   grade violation_code violation_description      count
##   <chr> <chr>          <chr>          <int>
## 1 A     10F          Non-food contact surface or equipment made of una... 5582
## 2 A     06D          Food contact surface not properly washed, rinsed ... 1802
## 3 A     06C          Food, supplies, or equipment not protected from p... 1749
## 4 A     10B          Anti-siphonage or back-flow prevention device not... 1712
## 5 A     02G          Cold TCS food item held above 41 °F; smoked or pr... 1300
## 6 A     02B          Hot TCS food item not held at or above 140 °F.      1247
## 7 A     08A          Establishment is not free of harborage or conditi... 997
## 8 A     10G          Dishwashing and ware washing: Cleaning and saniti... 619
## 9 A     04L          Evidence of mice or live mice in establishment's ... 528
## 10 A    06F          Wiping cloths not stored clean and dry, or in a s... 516
## # i 209 more rows
```

```
#grades w/violation code
restaurant_data %>%
  count(grade, violation_code) %>%
  arrange(grade, desc(n))
```

```
## # A tibble: 151 × 3
##   grade violation_code      n
##   <chr> <chr>          <int>
## 1 A     10F            5650
## 2 A     06D            1802
## 3 A     06C            1780
## 4 A     10B            1723
## 5 A     02G            1307
## 6 A     02B            1255
## 7 A     08A            1025
## 8 A     10G             636
## 9 A     04L             538
## 10 A    06F             520
## # i 141 more rows
```

```
#dataset summary
restaurant_data %>%
  summarise(
    Total_Records = n(),
    Restaurants = n_distinct(camis),
    Grades = n_distinct(grade),
    Violation_Codes = n_distinct(violation_code),
    Inspection_Dates = n_distinct(inspection_date)
  )
```

```
## # A tibble: 1 × 5
##   Total_Records Restaurants Grades Violation_Codes Inspection_Dates
##   <int>          <int> <int>          <int>          <int>
## 1      25026      25026     3             60            1192
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.